

Implementation Status (as of v0.4)

Code Inventory

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src/
__init__.py      # package
__main__.py      # entry: python -m src <command>
types.py         # 280 lines -- FirmState, QuarterFlows, Decisions, SimParams, CompustatRow
accounting.py    # 260 lines -- post_quarter() pure function, validation, Compustat builder
clamping.py      # 190 lines -- clamp_decisions() 6-step priority-order algorithm
demand.py        # 200 lines -- multinomial logit demand model (deterministic fallback)
config.py        # 110 lines -- YAML config + per-agent LLM config
llm_backends.py  # 160 lines -- OpenRouter + Ollama + Mock backends, JSON extraction
orchestrator.py  # 500 lines -- 9-phase quarter loop, DCF valuation, debt/equity mechanism
prompts.py       # 350 lines -- firm + environment prompts with personality archetypes
board_discussion.py # 300 lines -- 3-part management process (review/discuss/plan)
operational_reports.py # 300 lines -- R&D progress + brand/marketing reports
product_specs.py # 200 lines -- per-firm product sheet generator
memory.py        # 100 lines -- AgentMemory local accumulation
world_secrets.py # 300 lines -- hidden env context (templates + seed)
output_organizer.py # 200 lines -- structured output folders + cross-run DB
cli.py           # 380 lines -- run/smoke commands, agent wiring

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Total:           ~3,830 lines

tests/
test_accounting.py # 230 lines -- 42 tests (doc 16 golden fixture)
test_clamping.py   # 200 lines -- 15 edge cases (doc 17)
test_integration.py # 170 lines -- 7 integration tests (5Q + 20Q smoke)

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Total:           ~600 lines, 64 tests, all passing
```

Features Implemented

Core Simulation

- [x] Quarterly accounting cycle (IS, BS, CF, all invariants)
- [x] FIFO inventory, capacity utilization multiplier, process R&D COGS reduction
- [x] Feasibility clamping (6-step priority: COGS > mandatory > discretionary > payouts)
- [x] Multinomial logit demand model (deterministic fallback)
- [x] Firm default detection + slot tracking
- [x] Tax with NOL carryforward (80% usage limit)
- [x] Capability and brand stocks with diminishing returns (cap at 100)
- [x] R&D cumulative tracking toward generation thresholds
- [x] Compustat panel (~45 columns, Compustat-compatible naming)
- [x] Checkpoint-compatible WorldState

Agent System

- [x] Multi-step firm agent: board discussion (LLM call #1) then decision (LLM call #2)
- [x] Environment agent with narrative gazette generation
- [x] 5 strong personality archetypes (Aggressive Growth, Premium Innovator, Value Operator, Fast Follower, Marketing Powerhouse)
- [x] Board discussion: forecast review, CFO financing plan, COO ops plan, CEO strategy
- [x] Business plan forecasting (next-quarter targets stored and compared to actuals)

Financial System

- [x] DCF-based equity valuation (20Q projection, discount rate, terminal value, revenue multiple cross-check)
- [x] Auto-granted revolving credit (2x revenue, risk-adjusted rate)
- [x] Term debt issuance (up to 50% of assets, leverage-based pricing)
- [x] Equity secondary offerings (at market price with 5% discount)
- [x] Interest expense flows through IS (tax shield automatic)

Information Architecture

- [x] PUBLIC/PRIVATE separation enforced at orchestrator level
- [x] Per-firm info packages (no firm sees another's private data)
- [x] Environment omniscient (sees all private data + world secrets)
- [x] World secrets: hidden context with research paths, pre-planned events, firm-specific factors
- [x] Template categories + seed randomization for world secrets
- [x] Operational reports (R&D + brand) generated from state, shared with own firm + environment only

Memory System

- [x] AgentMemory with local accumulation (no redundant re-sends)
- [x] Board minutes persist across quarters (prior meeting referenced in next)
- [x] Forecast storage and review (plan vs actuals comparison)

Output System

- [x] Structured folders: public/, firms/firm_*/, environment/
- [x] Per-quarter: board minutes, product specs, gazettes
- [x] Cross-run database: compustat_all.csv + run_index.csv
- [x] World secrets saved to environment/ (never shared with firms)

LLM Configuration

- [x] Default: DeepSeek V3.2 via OpenRouter (~\$2.50/run)
- [x] Per-agent backend override (Ollama, OpenRouter, Anthropic, Mock)
- [x] Temperature: 0.3 for firms, 0.4 for environment
- [x] JSON extraction with retry (2 attempts + fallback)

Features NOT Yet Implemented

From Architecture Docs

- ☐ Private start mode (PE/VC funding, IPO transition)
- ☐ Equity Market agent (separate from built-in pricing)
- ☐ Investment Bank agent (advisory)
- ☐ Commercial Bank agent (LLM-driven credit decisions)
- ☐ Credit Fund agent (LLM-driven term debt)
- ☐ M&A with goodwill and impairment
- ☐ Leasing, stock compensation, workforce detail
- ☐ Entry/exit (new firms entering after default)
- ☐ Death-spiral prevention (slot pausing)
- ☐ Streamlit dashboard
- ☐ Cross-run learning (agents use past simulation data)
- ☐ Agent self-summarization for memory compression

Known Gaps

- Buybacks and dividends: clamping blocks dividends (negative RE) and LLMs don't request buybacks. Will activate when firms become profitable.
- Employee count (empq): always zero (workforce toggle OFF)
- Secondary equity: mechanism exists but LLMs rarely request it (board CFO recommends it but decision LLM doesn't always follow through)
- Compustat: sstkq and fincfq don't perfectly reconcile with balance sheet changes from Phase 7b financing (known approximation)

Test Results

Test	Count	Status
Accounting (doc 16 fixture)	30	PASS
Utilization multiplier	8	PASS
Tax/NOL	2	PASS
Edge cases (zero production, sell	2	PASS
Clamping (doc 17 edge cases)	15	PASS
Integration (5Q + 20Q smoke)	7	PASS
Total	64	ALL PASS

Run History

Run	Firms	Quarters	Key Observation
Pre-fix	3	5	R&D at \$10M minimum, all
Post-fix 1	3	5	R&D \$30M, prices still \$9
Post-fix 2	5	8	Firms adapt as cash deple
Board discussion	3	3	CFO presents financing op
Full restructure	3	5	firm_0 defaults Q4 (aggre